VENDOR PERFORMANCE ANALYSIS

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Abstract:

An essential component of supply chain management and business intelligence is the assessment of vendor performance. Vendors are crucial to modern businesses' overall profitability, cost optimization, and operational effectiveness. This study offers a thorough data-driven method for evaluating vendor performance using sophisticated analytical methods and visual aids. Key performance indicators like total sales, total purchases, gross profit, profit margin, and unsold capital are used by the proposed system to assess vendors. To offer real-time insights into vendor contribution and performance distribution, a dashboard-based analytical framework is created. To find high-performing suppliers, under-performing brands, and possible risk areas, the study uses methods like Pareto analysis, clustering, and risk segmentation. The findings support the Pareto principle by showing that a small number of vendors account for a sizable amount of overall revenue. Furthermore, a sizable portion of vendors fit the low-performance category, suggesting areas for improvement. Additionally, the system determines which brands need to make changes to their prices and promotions in order to boost sales. This study shows how data analytics can improve decision-making, boost vendor management tactics, and boost corporate profitability. For businesses looking to maximize their vendor ecosystem, the suggested framework offers a scalable and effective solution.

keywords:

Performance evaluation of vendors, data analytics, business intelligence, Pareto analysis, clustering, risk segmentation, data visualization, dashboard analytics, profit analysis, and KPI (Key Performance Indicators)

I. Introduction

Organizations must constantly optimize their operations to sustain profitability and efficiency in today's fiercely competitive business environment. Vendor management is one of the most important aspects of business operations. Vendors are in charge of providing goods and services, and the success of the business as a whole, delivery

schedules, and product quality are all directly impacted by their performance.

Conventional vendor evaluation techniques frequently fall short of providing a comprehensive picture of vendor performance because they rely on manual analysis and simple metrics. Large datasets cannot be handled by these methods because they are not scalable. As a result, companies struggle to find high-performing vendors, spot inefficiencies, and successfully manage risks.

Organizations can now examine vast amounts of vendor data to derive valuable insights thanks to the development of data analytics and business intelligence tools. Analyzing vendor performance entails assessing a number of factors, including risk levels, profit margins, purchase value, and sales contribution. Decision-makers can obtain a comprehensive understanding of vendor activities by combining these metrics into a single dashboard.

The goal of this project is to create a Vendor Performance Analysis system that uses data analytics methods to assess vendor performance. The system highlights areas for improvement, identifies low-performing vendors, and offers insights into vendor contribution. Additionally, it uses segmentation and clustering techniques to group vendors according to risk and performance.

II. Related Work

A study on the Vendor Performance Analytics Dashboard stresses the need to combine metrics for procurement, cost, and quality into one platform for making decisions in real time [1]. The study shows that old ways of evaluating vendors use manual reports and data that isn't all in one place, which slows down decision-making and makes things less efficient. The suggested system lets businesses keep an eye on how well their vendors are doing, cut costs, and make the supply chain more open by using business intelligence tools and data analytics. It also recommends using advanced analytics and machine learning to get better vendor evaluations and make predictions.

Another study on Vendor Management System using Machine Learning is about making a smart dashboard that can handle many vendors at once [2]. The system has features like tracking purchase orders, analysing sales, and reporting on vendor performance. It shows how machine learning can be used to find the best vendors and predict the quality of vendors based on past performance. The study also shows how important it is to use dashboards to visualise data. This makes it easier to understand complicated data and helps businesses make decisions faster and better.

Both studies stress how business intelligence dashboards can turn raw data into useful information [1][2]. Dashboards let you keep an eye on important performance indicators like sales, purchase costs, and vendor contributions in real time, which makes operations more efficient. The addition of machine learning also makes the system

better at doing predictive analysis and helping with strategic decision-making.

But current systems still have problems with things like processing large datasets, combining data from different sources, and making sure the analysis is correct [2]. By combining data analytics, machine learning, and visualisation techniques, you can get around these problems and find a better and more scalable way to analyse vendor performance.

This project builds on these research works by bringing together data processing, exploratory data analysis, machine learning models, and Power BI dashboards into one system.The proposed approach enables efficient vendor evaluation, better inventory management, and improved business decision-making through data-driven insights.

III. Existing System

The existing systems for vendor performance evaluation are primarily based on traditional and manual approaches that rely on limited data analysis techniques. In most organizations, vendor assessment is carried out using basic metrics such as purchase cost, delivery time, and product quality. These evaluations are often performed using spreadsheets or simple reporting tools, which lack the ability to process large volumes of data efficiently.

Conventional systems do not provide a comprehensive view of vendor performance, as they focus on isolated parameters rather than integrating multiple performance indicators such as sales, profit, and risk. As a result, decision-making becomes less accurate and may lead to inefficient vendor selection and resource allocation.

Furthermore, existing systems lack advanced analytical techniques such as the Pareto Principle, clustering, and risk segmentation. Without these methods, it becomes difficult to identify high-performing vendors, detect underperforming vendors, or analyze patterns in vendor behavior. The absence of predictive and analytical capabilities limits the effectiveness of vendor evaluation.

Another major limitation of current systems is the lack of proper data visualization. Most traditional approaches present data in tabular formats, making it difficult for decision-makers to interpret insights quickly. The absence of interactive dashboards and graphical representations reduces the efficiency of analysis and delays decision-making processes.

In addition, existing systems are not scalable and often require manual updates and maintenance. They do not support real-time analysis, which is essential for modern business environments where data changes frequently. This results in outdated insights and reduced responsiveness to market conditions.

IV. Proposed Methodology

The proposed methodology focuses on analyzing vendor performance using data analytics and machine learning. It starts with collecting and ingesting multiple datasets into a SQLite database, followed by data cleaning, preprocessing, and feature engineering to create key metrics like Gross Profit and Profit Margin. The data is then integrated into a unified dataset (vendor_sales_summary) and analyzed using EDA techniques to identify trends and patterns. Machine learning models such as Random Forest and XGBoost are applied, achieving an accuracy of 99.59%, and

finally, insights are visualized through interactive Power BI dashboards to support effective decision-making.

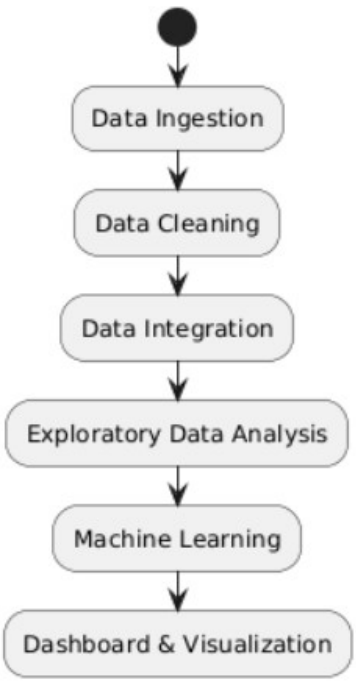


Figure Steps in proposed approach

1. Data Ingestion

Data ingestion is the first and one of the most important stages in the proposed system. It involves collecting, organizing, and storing large amounts of raw data from various sources into a structured format. In this project, we gather vendor-related data from multiple datasets and store it in a centralized SQLite database. This setup makes processing and analysis more efficient. Given the vast size of the datasets, which contain millions of records, we use a chunk-based ingestion approach to load the data in smaller portions. This method boosts performance, prevents memory issues, and ensures smooth handling of large-scale data.

1.1 Purchases Dataset

The purchases dataset is one of the main data sources for this project, holding about 2.37 million records. It includes details about purchase transactions such as inventory identifiers, store information, product descriptions, vendor details, and purchase order information. Additionally, it has financial details like purchase price, quantity, and total cost. This dataset is vital for understanding vendor supply patterns, procurement activities, and cost analysis.

1.2 Purchase Prices Dataset

The purchase prices dataset has over 12,000 records and provides detailed information about product prices from different vendors. It includes attributes like product brand, description, size, volume, and purchase price along with vendor details. This dataset is critical for analyzing price differences among vendors and finding cost-effective procurement strategies.

1.3 Vendor Invoice Dataset

The vendor invoice dataset contains around 5,500 records and includes transaction details between vendors and the organization. It has information like invoice dates, purchase order dates, payment dates, quantities, total transaction amounts, and shipping costs. This dataset helps in analyzing vendor payment cycles, additional costs, and overall financial interactions with vendors.

1.4 Beginning Inventory Dataset

The beginning inventory dataset contains over 200,000 records and shows the stock levels at the start of a specific period. It includes details such as inventory identifiers, store locations, product descriptions, stock availability, pricing, and start dates. This dataset helps us understand the initial inventory position and product availability.

1.5 End Inventory Dataset

The ending inventory dataset holds more than 220,000 records, representing stock levels at the end of a given period. It shares similar attributes with the beginning inventory dataset. By comparing the beginning and ending inventory datasets, we can track inventory movement and calculate important metrics such as stock turnover.

1.6 Sales Dataset

The sales dataset is the largest in this project, containing over 12.8 million records. It includes detailed information about sales transactions such as product details, sales quantity, sales revenue, sales price, and vendor information. This dataset is crucial for analyzing revenue generation, customer demand patterns, and the role of vendors in overall sales performance.

2. Data Preprocessing

Data preprocessing is a key step in the proposed system. It makes sure that the raw data collected during ingestion is clean, consistent, and ready for further analysis and modeling. Since the datasets are large and come from many sources, they may have inconsistencies like missing values, duplicates, incorrect formats, and noisy data. Therefore, we perform preprocessing to improve data quality and reliability.

3.Data Cleaning

Data cleaning means finding and correcting errors in the datasets. This includes dealing with missing values by removing or filling them, getting rid of duplicate records, and fixing inconsistent or invalid entries. We pay special attention to numerical fields such as purchase price, sales values, and quantities to ensure accuracy. We also identify and manage outliers carefully to prevent skewing the analysis results. This step ensures that the dataset is free from inconsistencies and is reliable for further processing.

4.Data Transformation

Data transformation changes the data into a suitable format for analysis. This includes converting date columns like purchase date, sales date, and payment date into proper datetime formats to enable time-series analysis. We standardize numerical fields as needed to keep consistency across datasets. We also standardize textual fields like vendor names, product descriptions, and store locations to avoid duplicates due to variations in naming conventions.

5.Feature Engineering

Feature engineering is important for improving the system's analytical capabilities. We create new features from existing data to provide deeper insights into vendor performance. We calculate metrics like Gross Profit by subtracting purchase cost from sales revenue, and we compute Profit Margin to measure profitability percentage. We derive Stock Turnover to see how efficiently we are using inventory, and we calculate the Sales-to-Purchase Ratio to evaluate the balance between demand and supply. These engineered features enhance both analysis and the performance of machine learning models.

6.Data Integration

Data integration combines multiple datasets into one unified dataset for analysis. In this project, we integrate datasets like purchases, sales, inventory, purchase prices, and vendor invoices using structured SQL queries. This step ensures that all relevant information related to vendor performance is brought together in one dataset.

The integrated dataset, known as the vendor sales summary, provides a complete view of

vendor activities, including procurement, sales performance, inventory movement, and financial metrics. By merging these datasets, we can effectively analyze the relationships between different variables. For example, we can study how purchase prices affect sales profitability or how inventory levels impact sales performance.

V. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to gain insights into the dataset and understand underlying patterns, trends, and relationships. This step is essential for identifying key characteristics of the data before applying machine learning models. Various statistical techniques and visualization methods are used during this phase.

The analysis includes generating summary statistics such as mean, median, and standard deviation to understand data distribution. Visualization techniques such as histograms, bar charts, and scatter plots are used to represent relationships between variables. Correlation analysis is conducted to identify dependencies between features such as sales, profit, and inventory levels.

Through EDA, important insights are derived, such as identifying top-performing vendors based on sales and profit, detecting low-performing vendors, and analyzing seasonal trends in sales. Outliers and anomalies are also identified during this stage, which helps in improving data quality and model performance.

VI. Machine Learning

Machine learning techniques are applied in this project to analyze vendor performance and generate predictive insights. Advanced algorithms such as Random Forest and XGBoost are selected due to their ability to handle large datasets and capture complex relationships between variables. These algorithms are particularly effective for structured data and provide high accuracy.

The dataset is divided into training and testing sets to evaluate model performance. The models are trained using historical data, allowing them to learn patterns and relationships between input features and target variables. During the evaluation phase, performance metrics such as accuracy are used to measure the effectiveness of the models.

Among the implemented models, the best-performing model achieved an accuracy of approximately 99.59 percent, indicating a high level of prediction reliability.

These machine learning models help in identifying key factors that influence vendor performance and enable predictive analysis, which can be used for future decision-making.

Algorithms:

A. XG BOOST:XGBoost is a more advanced version of gradient boosting that builds models one after the other to reduce errors. In this project, it is used to make predictions how efficiently a vendor will perform based on metrics like sales, the quantity of items they buy, the margin of profit, and how rapidly they move out of stock. The model is very accurate and performs efficiently, so it can deal with large datasets. Its performance makes it possible to make reliable decisions in real-world business applications.

B. **K-Means Clustering:** K-Means clustering is an approach of organising vendors into groups based on their performance levels into distinct groups. This helps you find patterns, like high, low, and medium-performing vendors. The clustering results give us useful information about how vendors behavior, which helps us make better decisions. In the end, it helps

businesses choose the best vendors and make more profitability and efficiently.

Accuracy

Accuracy tells you how correct the model is overall by figuring out what percentage of the predictions were correct.

What does accuracy mean?
Accuracy = (TP + TN) / (TP + TN + FP + FN)
Where:
TP (True Positive): correctly guessed positive cases
TN (True Negative): correctly guessed negative cases
FP (False Positive): cases that were wrongly thought to be positive

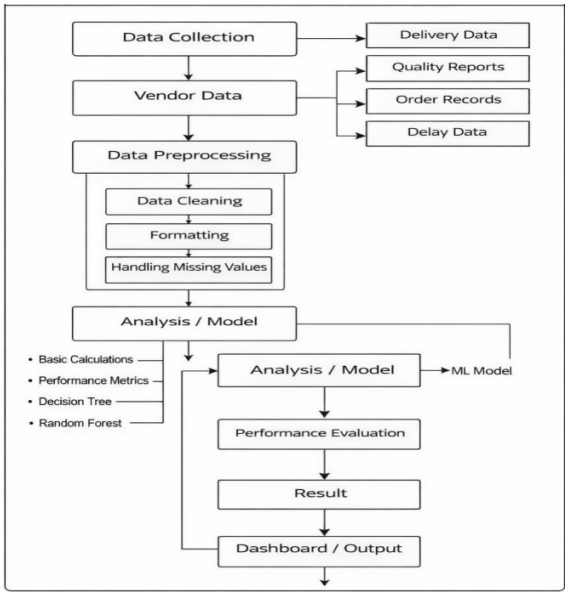
FN (False Negative): Incorrectly predicted cases that were not true

The implemented model got an overall validation accuracy of 99.59%.

VII. Architecture of the System

The system architecture of the proposed Vendor Performance Analytics System defines the structure and interaction of various components involved in data processing, analysis, and visualization. The system is designed using a layered approach that includes data ingestion, data processing, machine learning, and visualization layers. Each layer performs a specific function and interacts with other layers to ensure smooth data flow and efficient processing.

The architecture ensures scalability and performance by handling large datasets efficiently and enabling real-time analytical insights. The integration of machine learning models with visualization tools further enhances the system's capability to support intelligent decision-making.



VIII. Dashboard and Visualization

The final stage of the proposed methodology involves presenting the analyzed data and insights through interactive dashboards. Visualization plays a crucial role in making complex data understandable and accessible to users. In this project, Power BI is used to develop dynamic and interactive dashboards.

The dashboards provide visual representations of key performance indicators such as top vendors, profit margins, sales trends, inventory turnover, and vendor contribution. Users can interact with the dashboard by applying filters, drilling down into specific data points, and exploring trends over time.

visualizations help stakeholders quickly understand business performance and make informed decisions. For example, managers can identify high-performing vendors, detect inefficiencies, and optimize procurement strategies. The use of dashboards enhances the overall usability of the system and ensures that insights are effectively communicated.

IX. Results

The project's results give a complete understanding of how effectively vendors are performing by using large datasets and machine learning methods. It processes information about purchases, sales, inventory, and vendor transactions to deliver useful insights.

The system uses K-Means clustering and XGBoost to identify patterns in vendor performance, organise vendors into groups, and provide key performance indicators like profit margin, stock turnover, and sales trends. Interactive dashboards display the final results, which helps people to make better decisions.

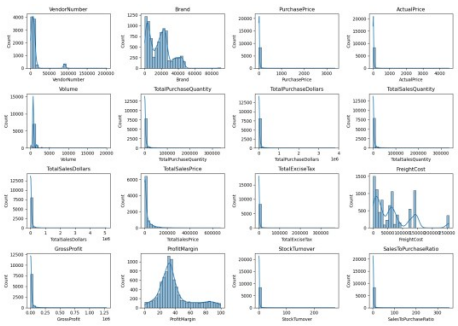


Figure 9.1 Distribution of Numerical Variables Using Histogram Plots

Figure 9.2 Vendor Cluster Based on Sales and Profit

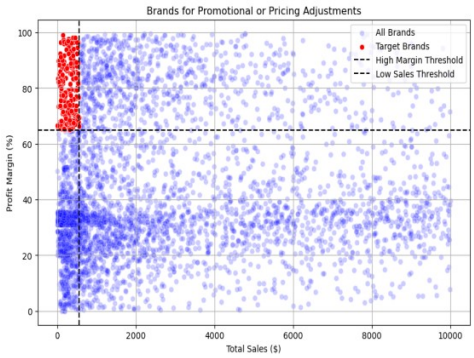


Fig. 9.3 Brands for Promotional or Pricing Adjustments

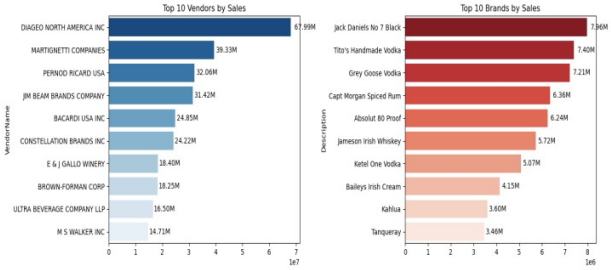


Fig 9.4 Top 10 Vendors and Brands by Sales

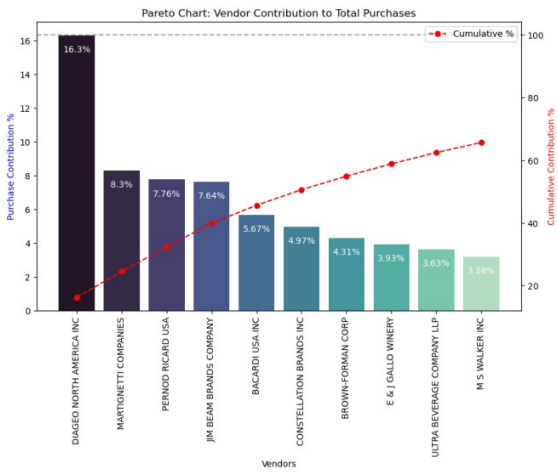


Fig 9.5 Pareto Chart: Vendor Contribution to Total Purchases

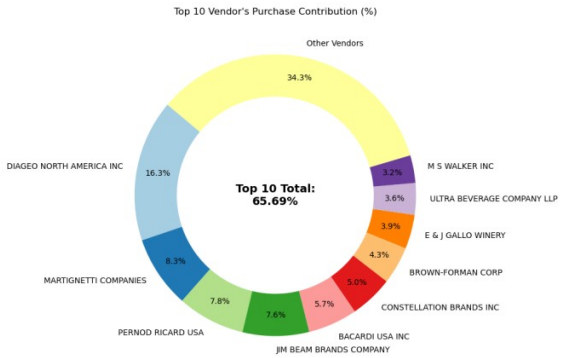


Fig 9.6 Top 10 Vendors Purchase contribution

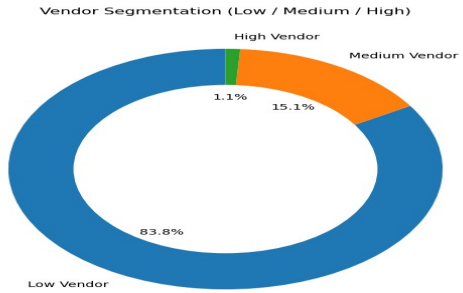


Fig 9.7 Vendor Segmentation)

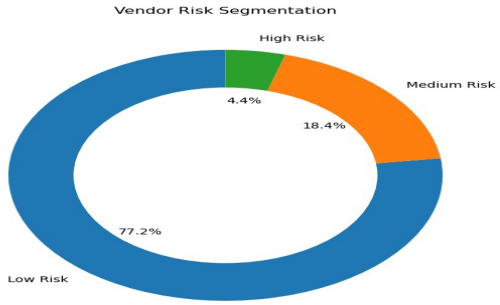


Fig 9.8 Vendor Risk Segmentation

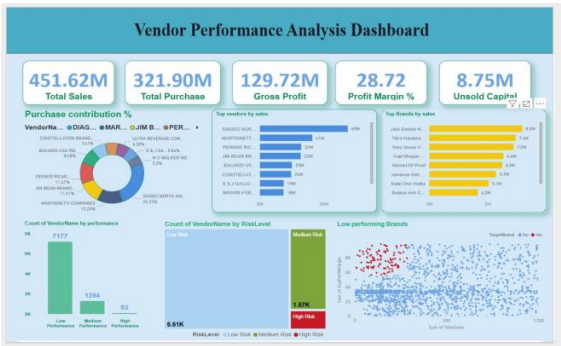


Fig 9.9 Vendor Performance Analysis Dashboard

X. Conclusion

The suggested vendor performance analytics system adopts a comprehensive and data-driven approach by using data analytics, machine learning, and visualization techniques to measure and improve vendor performance. The system uses a number of different datasets, like sales, purchases, inventory, and vendor information, Which are processed and turned into useful information through data preprocessing, feature engineering, and data integration. To make it easier to analyse vendors, key performance indicators like Gross Profit, Profit Margin, Stock Turnover, and Sales-to-Purchase Ratio are determined.

Clustering techniques like K-Means are used to classify vendors based on the level of performance and the level of risky they are. This helps to make it more simple to find high, low, and medium-performing vendors . Machine learning models are also used to identify patterns and help with predictive insights, with an accuracy rate of 99.59%. The addition of Power BI dashboards to the system makes it even better by providing users interactive and easy-to-understand visualizations of data like vendor segmentation, risk analysis, top vendors, and sales trends. Overall, the system provides a successfully transforming raw business data into actionable insights, supporting efficiency, making decisions, cost optimization, and improved oversight of vendors.

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